Influencing Factors and Evaluation of Supply Chain Resilience in Manufacturing Perspective

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Abstract

With the increasing complexity and uncertainty of global supply chains, supply chain resilience has become an important topic of interest for businesses and academia. Measuring the level of supply chain resilience helps to assess its ability to cope with external shocks and risks. This paper analyzes the recursive hierarchical structure of supply chain resilience using the ISM model, reveals the relationship and degree of dependence between each link, and evaluates the level of manufacturing supply chain resilience using the entropy weight -TOPSIS method. By evaluating the supply chain resilience level of the representative manufacturing enterprises in the past six years, we have gained insight into the characteristics of the manufacturing supply chain and the challenges faced by the manufacturing industry, aiming at proposing references for the supply chain crisis management mechanism and promoting the high-quality development of the manufacturing industry.

CCS Concepts

• **Social and professional topics → Transborder data flow**.

Keywords

Supply chain resilience, ISM model, Recursive hierarchy, Entropy weight -TOPSIS method, Crisis management mechanism, High-quality development of manufacturing industry

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1 Introduction

Global supply chains are booming in the 21st century, bringing new opportunities and challenges to enterprises. As a member of the World Trade Organization (WTO), China plays a key role in the global supply chain and is the world's top goods trading country and manufacturing giant. According to a McKinsey report, China's manufacturing sector contributes more than 35% [1] to the nation. However, the outbreak of the epidemic has affected global production and creation, which has led to a huge impact on the production and operation of many companies. The main manifestation of this is the dual pressure on the demand side and the supply side. As a result of the sharp decline in effective consumer demand, automobile manufacturing enterprises faced shrinking export orders and increased potential risk of capital chain breaks, while the supply shortage of automobile parts and core raw materials hindered production progress. This not only had a direct impact on production efficiency but also increased the uncertainty faced by enterprises. At the same time, the strict control of logistics during the epidemic also directly affected the delivery rate of orders, leading to a decline in logistics efficiency and a significant increase in operating costs. The current manufacturing enterprises' understanding of the supply chain is far from enough, and supply chain resilience emphasizes more on the ability to respond flexibly and sustain operations in the face of unexpected accidents. Therefore, establishing an evaluation index system to quantitatively assess supply chain resilience and summarize the lessons learned from the epidemic is crucial for enterprises to cope with future challenges, and is also a pending issue in supply chain resilience assessment research, which contributes to the objective evaluation of global supply chains. The research ideas of this paper are mainly shown in Figure 1.

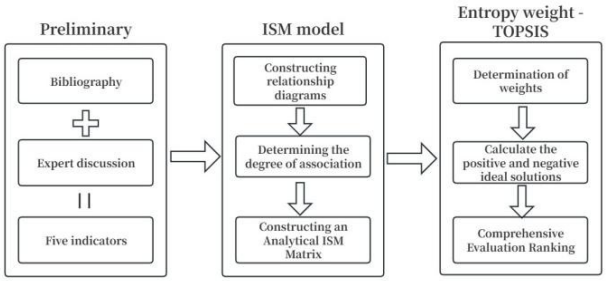

Figure 1: Roadmap for research.

2 LITERATURE REVIEW

With the ever-changing international situation, many scholars have recognized that improving supply chain resilience is conducive to mitigating supply chain disruptions (Naghshineh & Carvalho, 2022) [2]. Improving supply chain resilience has a very important position for enterprises to improve competitiveness (Shushan Zhang et al., 2023) [3]. Gaurav Kumar Badhotiya et al. [2022], studied the impact

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of overcoming the impact of disruptions on the resilience of the supply chain, and established an ISM model to analyze the relationship between the impacts of various indexes of supply chain resilience; Fan Xuemei et al [4]. used ISM model to analyze the structure of factors influencing the supply chain resilience of enterprises from the five dimensions of supply chain resilience.five dimensions, using the ISM model to analyze the structure of the factors influencing the supply chain resilience of enterprises; Wang Hongchun et al [5] also determine the indicators affecting supply chain resilience through literature research, and expert interviews, and construct the DEMATEL-ISM comprehensive model to analyze the degree of importance and the size of the influence of each factor on the supply chain resilience of assembly construction. The technique of sequential preference based on ideal solution similarity (TOPSIS) [6] is a classical multi-attribute decision-making method, which is widely used in decision-making or evaluation in various fields. The entropy method (EM) is often used to determine the attribute weights of TOPSIS, and the weights determined by EM are always called entropy weights (EW). The TOPSIS of attribute weights determined by entropy weight EM is called entropy-based TOPSIS [7]. E-TOPSIS is widely used in various fields of decision-making or assessment, such as port logistics function evaluation [8], livelihood indicator evaluation [9], urban carbon emission market trading risk assessment [10], and a comprehensive evaluation of the modern service industry [11].

3 RESEARCH METHODOLOGY

3.1 Supply Chain Resilience Indicator Construction

Tables By reviewing the domestic and international literature on the influencing factors of supply chain resilience, combined with the connotation of supply chain resilience, this paper argues that the influencing factors of supply chain resilience mainly include five dimensions: supply chain forecasting capability, supply chain response capability, supply chain adaptive capability, supply chain resilience capability and supply chain learning capability. (1)Supply chain forecasting capability: including risk awareness and production capability, the former refers to early warning, planning, assessment, risk avoidance, and control, and the latter refers to the timely adjustment of production and supply activities under changes and risks. (2) Supply chain responsiveness: includes flexibility and agility. Flexibility refers to the elasticity and adaptability of the supply chain, enabling enterprises to flexibly adjust resources; agility refers to the ability of the supply chain to quickly respond to risks and changes. (3) Supply chain adaptability: the retention of redundant resources is key. Redundant resources include alternate suppliers, alternate production equipment, etc. to ensure the continuity and stability of the supply chain. (4) Supply chain resilience: It includes financial strength, logistics support, contingency plans, and supply chain structure. Financial strength provides financial support, logistics support guarantees goods distribution and transportation, a contingency plan establishes an emergency response mechanism, and the complexity of the supply chain structure affects the degree of impact. (5) Supply chain learning ability: including organizational learning, knowledge creation, scientific research input, scientific research strength, and knowledge conversion. These capabilities help enterprises summarize experience, discover new knowledge, and improve supply chain capability and innovation.

This paper invites experts in the industry to identify and supplement based on the literature review, and after many discussions, finally obtains the index system of supply chain resilience influencing factors which contains 5 first-level indicators and 14 second-level indicators, as shown in Table 1.

3.2 ISM modeling

Interpretive Structural Modeling (ISM) is a method used to determine the interrelationships between various indicators. It is a systematic analytical method for revealing the interrelationships and hierarchical structure of factors in a complex system. It can help researchers clarify cause and effect relationships, analyze the primary and secondary relationships in a system, and provide a reliable basis for decision-making.ISM modeling can present these relationships as a hierarchical diagram that demonstrates the interactions and effects of the factors.

The main steps of the model are as follows:

3.2.1 Factor Determination. Determine the factors, through reviewing the domestic and international literature on the influence factors of supply chain resilience, combined with the connotation of supply chain resilience, this paper identifies five aspects of supply chain resilience as the factors to be analysed, i.e., predictive ability, responsiveness, adaptive ability, resilience and learning ability.

3.2.2 Expert Interviews. The experience of experts in supply chain related fields is collected. Then, based on the experts' opinions and judgments, a relationship graph between factors is constructed, in which each factor is represented as a node and the relationship between factors is represented by directed edges.

3.2.3 Determination of the degree of factor association. By analyzing the factors in the diagram step-by-step, the causal relationships between the factors are identified. In the first step, feedback loops identify possible circular dependencies between factors that may lead to system instability. In the second step, hierarchies are created by hierarchizing the factors according to their importance to form a hierarchical structure that reflects the relative degree of influence between the factors. In the third step, the adjacency matrix is established, based on the constructed index system of supply chain resilience, the adjacency matrix A is established, and the formula (1) for taking the values of the elements in matrix A are as follows:

$$a_{ij} = \begin{cases} 1, & \text{Influence factor } F_i \text{affects} F_j \\ 0, & \text{Influence factor } F_i \text{ has no impacton } F_j \end{cases} \tag{1}$$

3.2.4 ISM matrix construction. According to the above table the average value of each expert score, combined with the supply chain resilience factors for a series of correlation analyses, to obtain the neighboring matrix. In the appeal of the construction of the neighboring matrix on the use of Boolean matrix arithmetic rules for power operations, until to meet the (2) to obtain the reachable matrix (3).

$$(A+I)i-1 \neq (A+I)i = (A+I)i+1 \tag{2}$$

Table 1: Supply chain resilience influencing factors.

Level 1 indicators	Labels	Secondary indicators	Assessment methodology
Forecasting capacity F1	F11	Production capacity	Number of producers
	F12	Risk Awareness	Number of people
Reactivity F2	F21	Flexible	Inventory turnover
	F22	Agility	Production and marketing speed
Adaptability F3	F31	Redundant	Number of inventories
Resilience F4	F41	Financial strength	sales margin
	F42	Operating capacity	current ratio
	F43	Emergency response capacity	Inventory turnover
	F44	Supply chain structure	Percentage of total annual purchases by the top five vendors combined
Learning capacity F5	F51	Organizational learning capacity	Advanced training/trip
	F52	Knowledge creation capacity	Number of patents
	F53	Research investment	R&D investment
	F54	Research Strength	R&D staff
	F55	Knowledge transfer capacity	R&D investment conversion rate

$$M = (A + I)i \tag{3}$$

3.2.5 Analytical ism matrix. By calculating and analyzing the ISM matrix, the structural and hierarchical relationships between supply chain resilience factors can be derived and the hierarchy of factors can be determined as a means of assessing the relative importance between different supply chain resilience factors.

3.3 Entropy weight-Topsis model

The entropy weight-Topsis model fully combines the advantages of both the entropy weight method and the Topsis method of evaluation. The entropy weight method solves the problems of subjective weights and insufficient quantitative analysis in the traditional method, avoids over-concentration of weights, and provides objectivity for the evaluation; whereas the Topsis method can accurately reflect the gaps between the evaluation schemes, and derive the comprehensive index value by calculating the distance to the positive ideal solution and the negative ideal solution, and ultimately determines the best decision-making scheme.

The attributes to be evaluated acquire initial data by the evaluation indexes and establish a decision matrix X,x which is calculated as shown in (4).

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \tag{4}$$

In the above equation $Z_{ij}(i = 1, 2, \ldots, n; j = 1, 2, \ldots, m)$, idenotes the number of evaluation objects, Indicator value of the jth evaluation indicator, n is the evaluation object; m is the evaluation indicator.

3.3.1 Normalization. The data outline of each indicator is different, first normalize the indicator data to get the P matrix, calculated as shown in (5).

$$P_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^2}}, i = 1, 2, \ldots, m; j = 1, 2, \ldots, n \tag{5}$$

3.3.2 Entropy calculations. The entropy weight method [12] calculates the entropy value, the entropy value e_j of the jth evaluation index, calculated as shown in (6).

$$\mathrm{e}_j = -\frac{1}{\ln m} \sum_{i=1}^{m} p_{ij} \ln p_{ij}, i = 1, 2, \ldots, m; j = 1, 2, \ldots, n \tag{6}$$

3.3.3 Calculation of indicator weights. The entropy value of the jth evaluation metric, computed W_j methods (7):

$$W_j = \frac{(1 - e_j)}{\sum_{j=1}^{n} (1 - e_j)}, j = 1, 2, \ldots, n \tag{7}$$

3.3.4 Calculation of weighted decision matrix. Introduce the weight vector W into the normalization matrix X to obtain the normalized weighted decision matrix Z (8).

$$\mathrm{Z} = \begin{bmatrix} \mathrm{W_1P_{11}} & \mathrm{W_2P_{12}} & \cdots & \mathrm{W_mP_{1\,m}} \\ \mathrm{W_1P_{22}} & \mathrm{W_jP_{ij}} & \ldots & \mathrm{W_mP_{2\,m}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{W_1P_{n1}} & \mathrm{W_2P_{n2}} & \cdots & \mathrm{W_mP_{nm}} \end{bmatrix} \tag{8}$$

3.3.5 Calculation of Euclidean distances. Setting up positive (9) and negative (10) ideal solutions provides a comparison reference for each metric, which is used to assess how close the candidate solution is to the ideal state [13].

$$\begin{aligned} \mathrm{Z}^+ = \left(\mathrm{Z}_1^+, \mathrm{Z}_2^+, \ldots, \mathrm{Z}_\mathrm{m}^+\right) = \max\left(\{\mathrm{z}_{11}, \mathrm{z}_{21}, \ldots, \mathrm{z}_{\mathrm{n}1}\}, \right. \\ \left. \max\{\mathrm{z}_{12}, \mathrm{z}_{22}, \ldots, \mathrm{z}_{\mathrm{n}2}\}, \max\{\mathrm{z}_{1\,\mathrm{m}}, \mathrm{z}_{2\,\mathrm{m}}, \ldots \mathrm{z}_{\mathrm{nm}}\}\right. \end{aligned} \tag{9}$$

$$\begin{aligned} Z^- = \left(Z_1^-, Z_2^-, \ldots, Z_m^-\right) = \min\left(\{z_{11}, z_{21}, \ldots, z_{n1}\}, \right. \\ \left. \min\{z_{12}, z_{22}, \ldots, z_{n2}\}, \min\{z_{1m}, z_{2m}, \ldots z_{nm}\}\right. \end{aligned} \tag{10}$$

3.4 Calculation of positive and negative ideal solutions.

Calculate the positive and negative ideal solutions: use the Euclidean Distance to measure the difference between the solution and the positive and negative ideal solutions. Positive Ideal Solution Z^+ The Euclidean Distance between the positive ideal solution and the negative ideal solution Z^- The Euclidean Distance between the positive ideal solution and the negative ideal solution is calculated as follows:

Define the distance of the ith $(i = 1, 2, \ldots, n)$ distance of the first evaluation object from the maximum value (11).

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \left(Z_j^+ - z_{ij}\right)^2} \tag{11}$$

Define the distance of the ith (i=1,2,...,n) distance of the first evaluated object from the minimum (12).

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \left(Z_j^- - z_{ij}\right)^2} \tag{12}$$

3.5 Calculation of relative proximity.

Calculation of relative proximity (13): By calculating proximity C_i The object of judgment is analyzed, when C_i The higher and closer to 1, the closer to the positive ideal solution, the better the performance. The higher and closer to 1, the closer to the positive ideal solution, the better the performance.

$$C_i = \frac{D_i^-}{D_i^+ + D_i} \tag{13}$$

3.6 Calculation of indicator weights.

Can be calculated to pay the ith $(i = 1, 2, \ldots, n)$ un-normalized score for the first evaluation object. We normalize the final score by scaling the data to values in the range [0,1] by a certain percentage. (y denotes the original data, y' denotes the transformed indicator value $y_{\max}$ and $y_{\min}$ are the minimum and maximum values in the original dataset, respectively,the calculation formula (14) is as follows:

$$y' = (y - y_{\min}) / (y_{\max} - y) \tag{14}$$

4 EMPIRICAL RESEARCH

4.1 Data sources

The study adopts a small to large research approach, first taking Midea Group as a representative enterprise, and reflecting supply chain resilience through its development process over the years in depth. Later, it compares the three major representative home appliance companies in China with the three major representative home appliance companies in Japan to reflect what are the main factors influencing the strength of supply chain resilience. Meanwhile, multiple indicators covering the reliability, flexibility, adaptability, and synergy of the supply chain are used to comprehensively assess the level of supply chain resilience. Our data sample for this collection is the annual data of Midea Group from 2017-2022 and Gree Electric Appliances, Haier Smart, Panasonic, Hitachi, and Philips for 2022, with a total of 254 rows of records in 14 categories. The applied data are mainly extracted from the information of financial statements, sustainability reports, annual reports, and news reports on the official websites of the major companies.

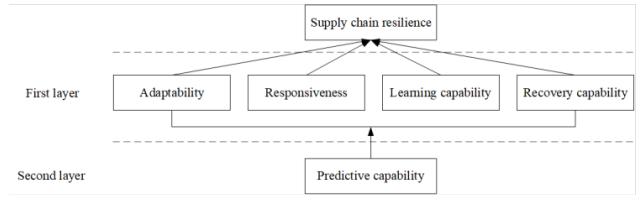

Figure 2: Supply chain resilience diagram.

4.2 Analysis of experimental results

4.2.1 Neighborhood matrix of factors affecting supply chain resilience. In this paper, we invited 11 professors from schools in this field and staff from various companies to score the ratings and combine them with the analysis of the influencing factors of supply chain resilience to obtain the adjacency matrix, as shown in Table 2.

4.2.2 ISM model Accessibility Matrix. Based on the 'adjacency matrix's basis, the sum of the 'adjacency matrix' and the 'unit matrix' is computed, which is used to compute the 'Accessibility Matrix' in the next step '. MATLAB is used to assist in the calculation of the relevant 'Accessibility Matrix' and analyze the relationship between the factors affecting the toughness of the supply chain. The accessibility matrix is shown in Table 3.

4.2.3 Hierarchical analysis tables. Using the "Accessibility Matrix" to analyze the correlation of each influence in the supply chain resilience factors, and using the hierarchical decomposition table to show the hierarchical distribution of each element, we get the progressive explanatory structure model of the supply chain resilience influencing factors. As shown in Figure 2. The model is then divided into three levels, i.e., surface direct influences, middle indirect influences, and deep root influences [14]. According to the results, it can be concluded that the first layer of responsiveness, resilience, learning ability, and adaptability directly affect the supply chain resilience of an organization, while the second layer of predictive ability indirectly affects the supply chain resilience through the first layer.

4.2.4 Analysis of the results of the supply chain resilience evaluation model.

Evaluation of the results of Midea Group's indicator weights. According to Figure 3, Midea's supply chain resilience has fluctuated in recent years, with low ratings in 2017 and 2018, improving in 2019, highest in 2020, declining in 2021, and improving in 2022. In combination with those shown in Table 4, training headcount, inventory count, and R&D investment/patents are the top three weighted metrics. The number of people trained reflects the company's focus on employee training to improve risk awareness; the

Table 2: Adjacency matrix.

	Forecasting capabilities	Responsiveness	Adaptive	Resilience	Learning ability	Supply chain resilience
Predictive capacity	0	1	1	1	1	0
Responsiveness	0	0	0	0	0	1
Adaptive	0	0	0	0	0	1
Resilience	0	0	0	0	0	1
Learning ability	0	0	0	0	0	1
Supply chain resilience	0	0	0	0	0	0

Table 3: Accessibility Matrix.

	Forecasting capabilities	Responsiveness	Adaptive	Resilience	Learning Learning ability
Forecasting capabilities	1	1	1	1	1
responsiveness	0	1	0	0	0
Adaptive	0	0	0	0	0
Resilience	0	0	1	1	0
Learning ability	0	0	0	0	1
Supply chain resilience	0	0	0	0	0

Table 4: Weighting table for Midea GROUP'S COMPOSITE indicators.

Level 1 indicators	Labels	Secondary indicators	weights
Predictive capacity F1	F11	Production capacity	0.0068
(0.3528)	F12	Risk Awareness	0.3460
Reactivity F2	F21	Flexible	0.0013
(0.0014)	F22	Agility	0.0001
Adaptability F3 (0.3043)	F31	Redundant	0.3043
	F41	Financial strength	0.0005
Resilience F4 (0.0089)	F42	Operating capacity	0.0031
	F43	Emergency response capacity	0.0013
	F44	Supply chain structure	0.0040
	F51	Organizational learning capacity	0.0756
	F52	Knowledge creation capacity	0.0094
Learning capacity F5 (0.3326)	F53	Research inputs	0.0045
	F54	Research Strength	0.0149
	F55	Knowledge transfer capacity	0.2282

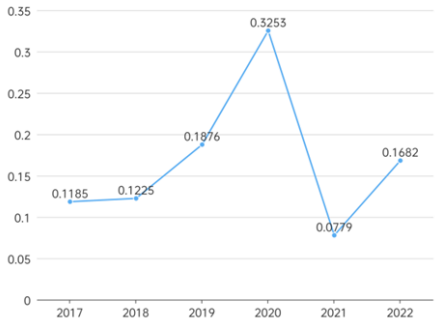

Figure 3: Midea Group Supply Chain Resilience, 2017-2022.

number of inventories reflects the importance placed on inventory management to ensure supply chain flexibility; and the R&D investment conversion rate reflects the importance of scientific and technological development and the level of technology. The comprehensive indicators show that Midea has made efforts to improve supply chain resilience, but there are still fluctuations. 2020 responded to the new crown epidemic by taking measures such as increasing inventory and adjusting production to gain a competitive advantage. 2021 faced supply chain disruptions, raw material shortages, and other issues, resulting in a decline in resilience. As the global outbreak is brought under control in 2022, the supply chain gradually stabilizes.

Evaluation of the results of the weighting of the six corporate indicators. As seen in Figure 4, the entropy weight-TOPSIS model evaluates the supply chain resilience of six major home appliance manufacturing companies, with Hitachi at the top of the list with the highest score. Considering the five indicators of prediction, reaction,

Table 5: Result of the weighting of indicators for the six largest companies.

Level 1 indicators	Labels	Secondary indicators	weights
Predictive capacity F1	F11	Production capacity	0.1126
(0.3528)	F12	Risk Awareness	0.0712
Reactivity F2	F21	Flexible	0.0115
(0.0014)	F22	Agility	0.0052
Adaptability F3	F31	Redundant	0.1592
	F41	Financial strength	0.0210
Resilience F4	F42	Operating capacity	0.0195
	F43	Emergency response capacity	0.0098
	F44	Supply chain structure	0.0246
	F51	Organizational learning capacity	0.2198
	F52	Knowledge creation capacity	0.0578
Learning capacity F5	F53	Research inputs	0.0991
	F54	Research Strength	0.1176
	F55	Knowledge transfer capacity	0.0710

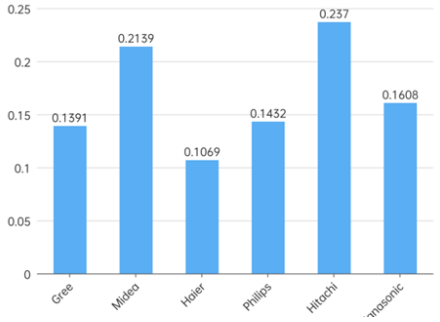

Figure 4: Supply Chain Resilience Score for Large Companies.

adaptation, recovery, and learning, the different weights reflect the different focuses of the companies on supply chain resilience.

4.3 Analysis of conclusions

Combined with Table 5, it can be analyzed:Organizational learning capability (F51) in learning capability (F5) has the highest weight, and improving the professionalism and skill level of employees through advanced training can help improve supply chain resilience. (2)Adaptability (F3) has the next highest weighting of inventory count, and a reasonable inventory level can safeguard supply chain flexibility. (3)Organizational learning ability (F5) has the next highest weight of research strength (F54), and strong research strength helps to improve enterprise competitiveness.(4)The production capacity (F11) in Forecasting Capability (F1) has a relatively high weighting of the number of producers, and increasing production capacity helps to cope with fluctuations in market demand and improve production efficiency. (5) Based on the comprehensive evaluation results, to improve supply chain resilience, the company can start by optimizing the production process, increasing production capacity, strengthening staff training, increasing investment in scientific research and scientific inventory management, as well as establishing a comprehensive emergency plan and lowering the risk level. These measures will help companies better adapt to market changes and demands and improve supply chain resilience.

5 CONCLUSIONS

By analyzing representative manufacturing enterprises in multiple countries, we can provide a new entry point for exploring the effects of supply chain resilience and improving the quality of manufacturing enterprise development. We can determine where supply chain resilience needs to be improved. This will help strengthen the resilience and adaptability of supply chains, improve productivity and quality, reduce costs, and better adapt to market demand and respond to risks. Below are our recommendations and countermeasures based on this study:

(1) Precise Inventory Management: Implement data-driven decision-making and regularly review inventory strategies to meet optimal inventory demands, thereby enhancing supply chain resilience. Inventory redundancy plays a significant role in the evaluation of supply chain resilience both domestically and internationally. Therefore, focus on data-driven decisions, precise forecasting based on market demand, and continuous optimization of inventory management.

(2) Enhancing Technological Innovation and Digitization: Increase investment in research and development, introduce smart manufacturing, the Internet of Things (IoT), and big data analytics to improve production efficiency, reduce costs, and promote digital supply chain management. Technological innovation capabilities are crucial in the evaluation of supply chain resilience globally. Digital transformation can enhance supply chain flexibility and responsiveness, fostering information sharing and collaboration.

(3) Establishing Emergency Management Mechanisms: Develop emergency response plans, crisis management processes, and business continuity plans to promptly address emergencies and mitigate uncertainties and risks through risk assessment and early warning mechanisms. The outbreak of the pandemic has underscored the importance of emergency management mechanisms. Enterprises need to establish comprehensive crisis management processes to handle various emergencies.

(4) Increasing Advanced Training Sessions: Emphasize advanced training to cultivate high-level talent and build teams with supply chain thinking and capabilities, thereby enhancing overall supply chain resilience. The frequency of advanced training sessions holds significant weight in the evaluation of supply chain resilience. Developing high-quality talent helps enterprises adapt to rapidly changing technologies and market environments, improving supply chain resilience.

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